

DEVELOPMENT ENGINEERING PROJECT

Decentralised Biogas Production with Centralised Upgrading & Grid Injection: When Is It Optimal?

Based on: 2014 BAB Hengeveld

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Pre Mid-Term

Overview of Research Paper

Studied the biogas production and upgrading system.

Understood the decentralized digester and central hub concept.

Work done till mid-sem:

- Studied the research paper and understood the full biogas-to-grid system.
- Analyzed the decentralized digester and centralized upgrading concept.
- Broke the project into transport, digestion, upgrading, injection, and validation sections.
- Developed the basic solution method for pipeline transport and central gas upgrading.

Pre Mid-Term

Methodology and Results

Pre-midsem, cost and energy validation were completed. Our Excel model was compared with the paper values.

Calculated results:

- Total production cost from paper: 71.2 €ct/m³.
- Total production cost from our Excel model: 49.8 €ct/m³.

Energy results:

- Total energy consumption from paper: 4.79 MJ/m³.
- Total energy consumption from our Excel model: 3.40 MJ/m³.

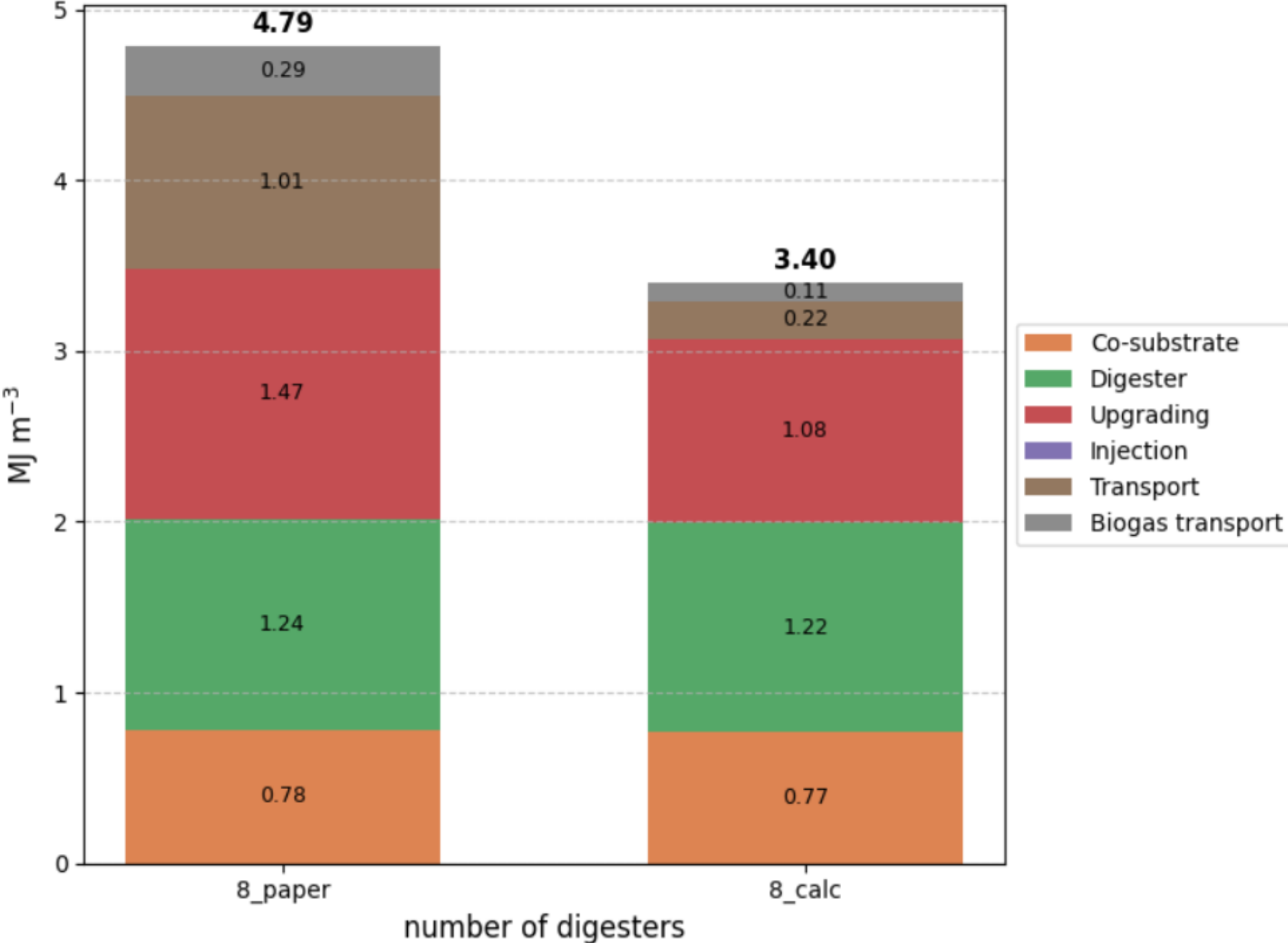
Additional work done:

- Calculated farm radius, transport distance, pipeline length, and compressor requirements.
- Cost and energy trends matched the reference paper closely.

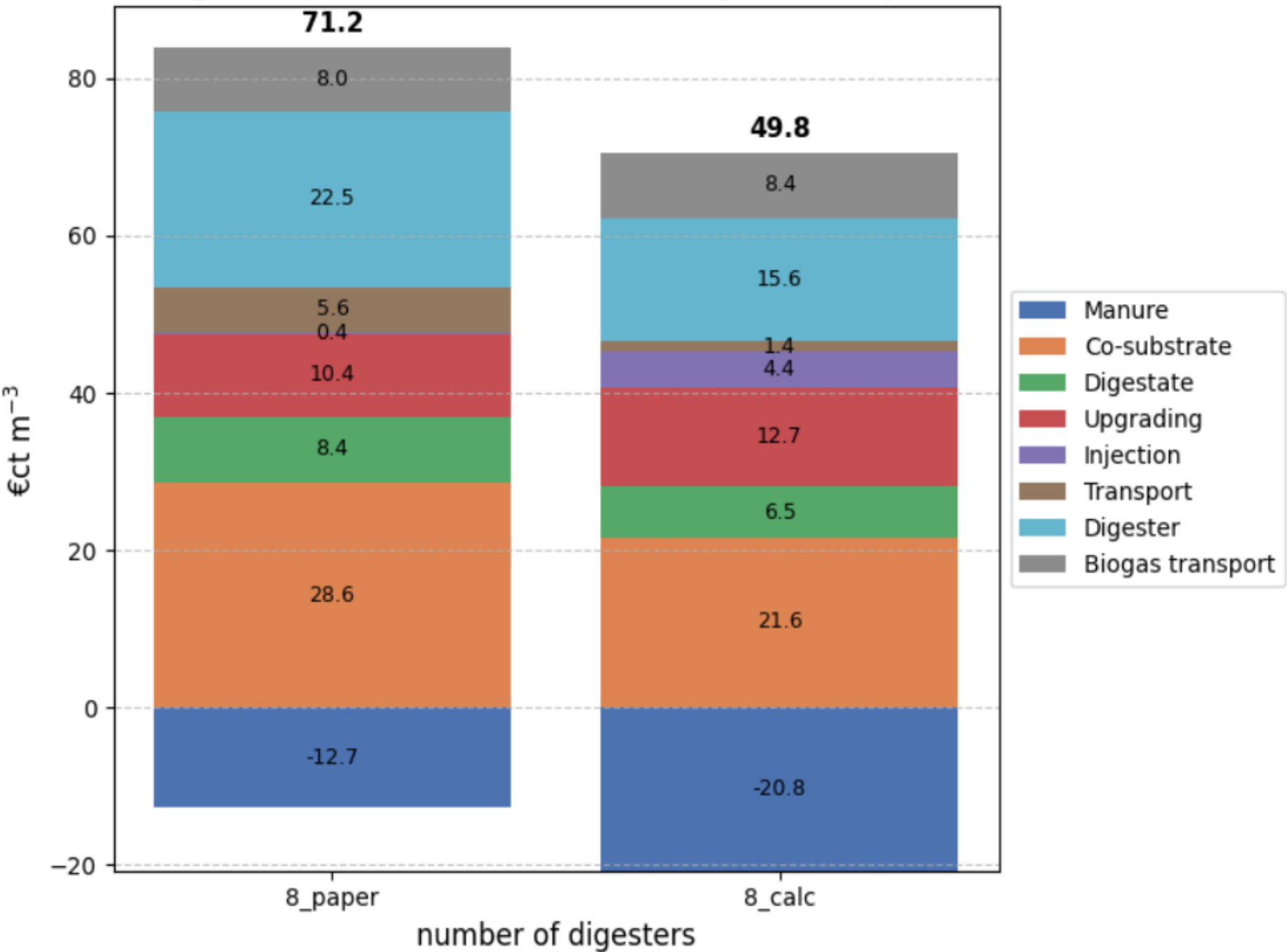
Pre Mid-Term

Results

Green gas energy use structure for decentralized digester configurations



Green gas cost structure for decentralized digester configurations



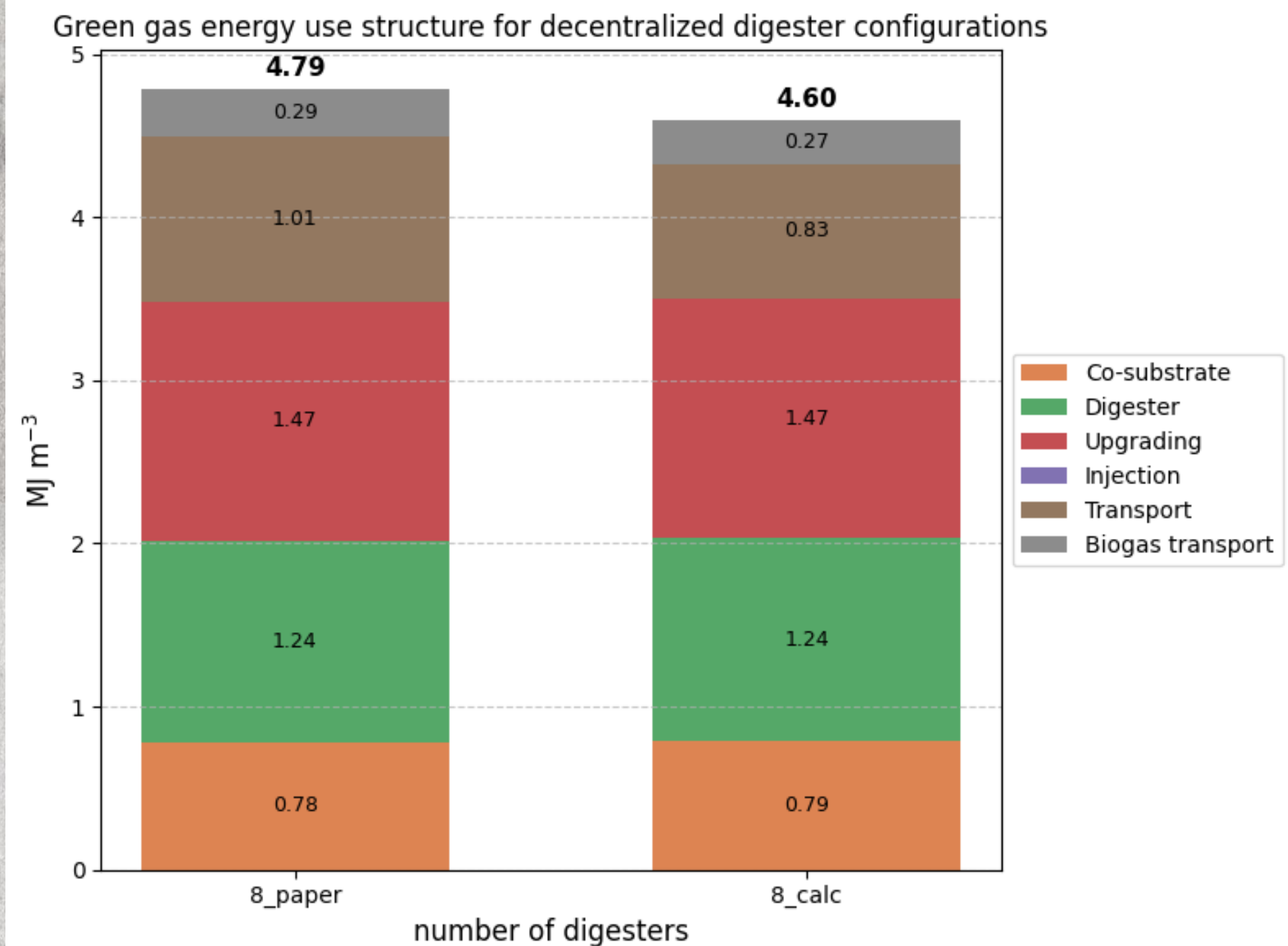
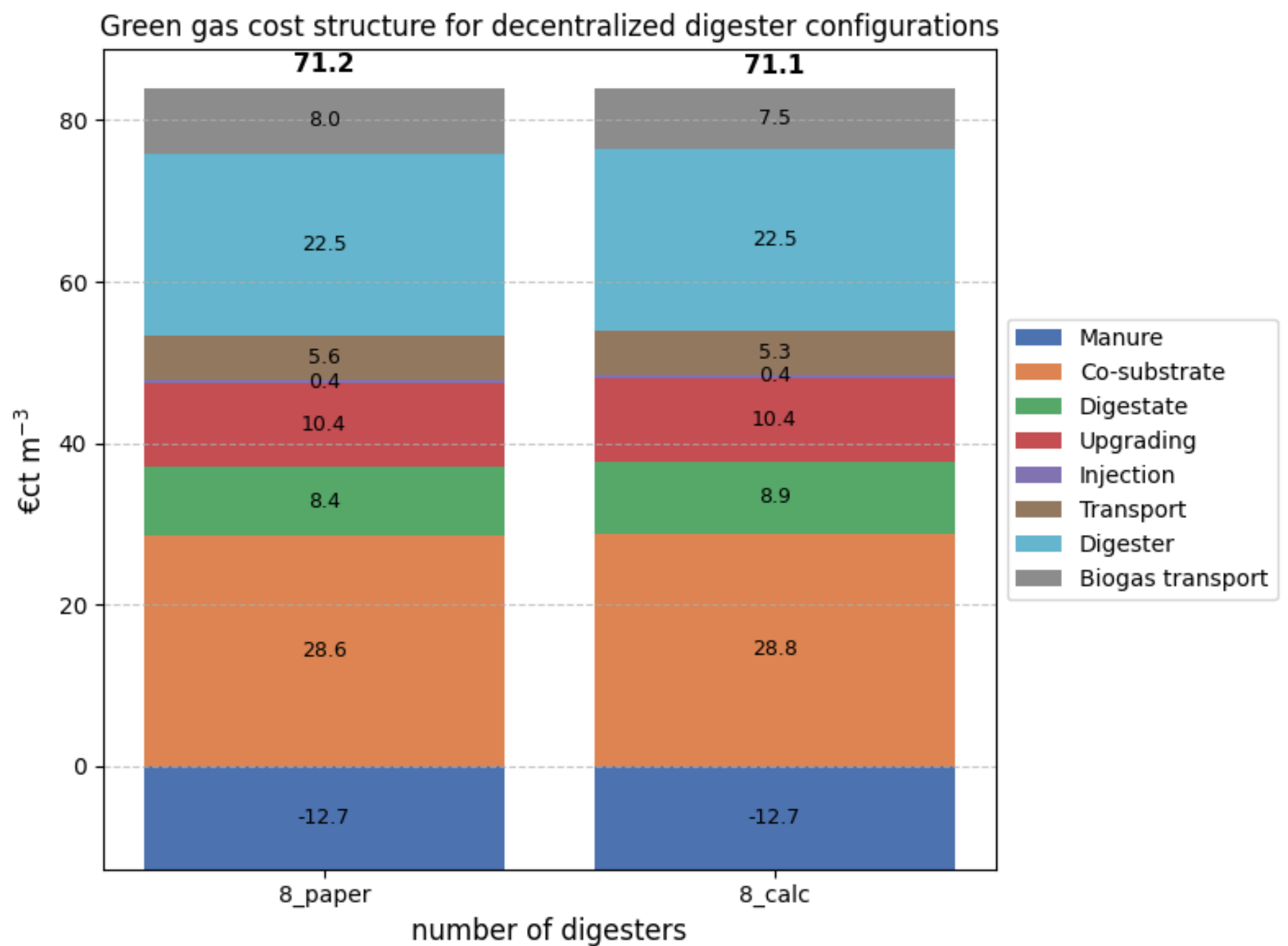
Post Mid-Term

Improvements made on Mid-term

- Initially, the cost and energy outputs from our Excel model did not match the values reported in the research paper.
- Conducted an in-depth review of additional research papers to identify accurate parameter values and modeling approaches.
- Refined key input parameters (e.g., No. of cows, Co-substrate, Co-substrate price, Investment of upgrading unit and digester) based on literature findings.
- Introduced justified assumptions for parameters where direct data was unavailable.
- Iteratively updated and calibrated the model to minimize deviation from reference results.
- Achieved close alignment of cost and energy outputs with the research paper, improving model accuracy and reliability.

Post Mid-Term

Results after changing parametric values



Implementation of the model

Selection of Study Location

Understand how geography and resource distribution affect biogas production systems.

Kota & Bikaner

Capture variation in:

- Biomass availability (manure + crop residue)
- Transport distances.

Updated key Parameters

Number of cattle:

- Kota:5149
- Bikaner:13050

Maize Production:

Transport distance

Number of digesters

Impact of Updated Parameters

Biomass production.

Green gas production

Transport cost

Energy consumption

References

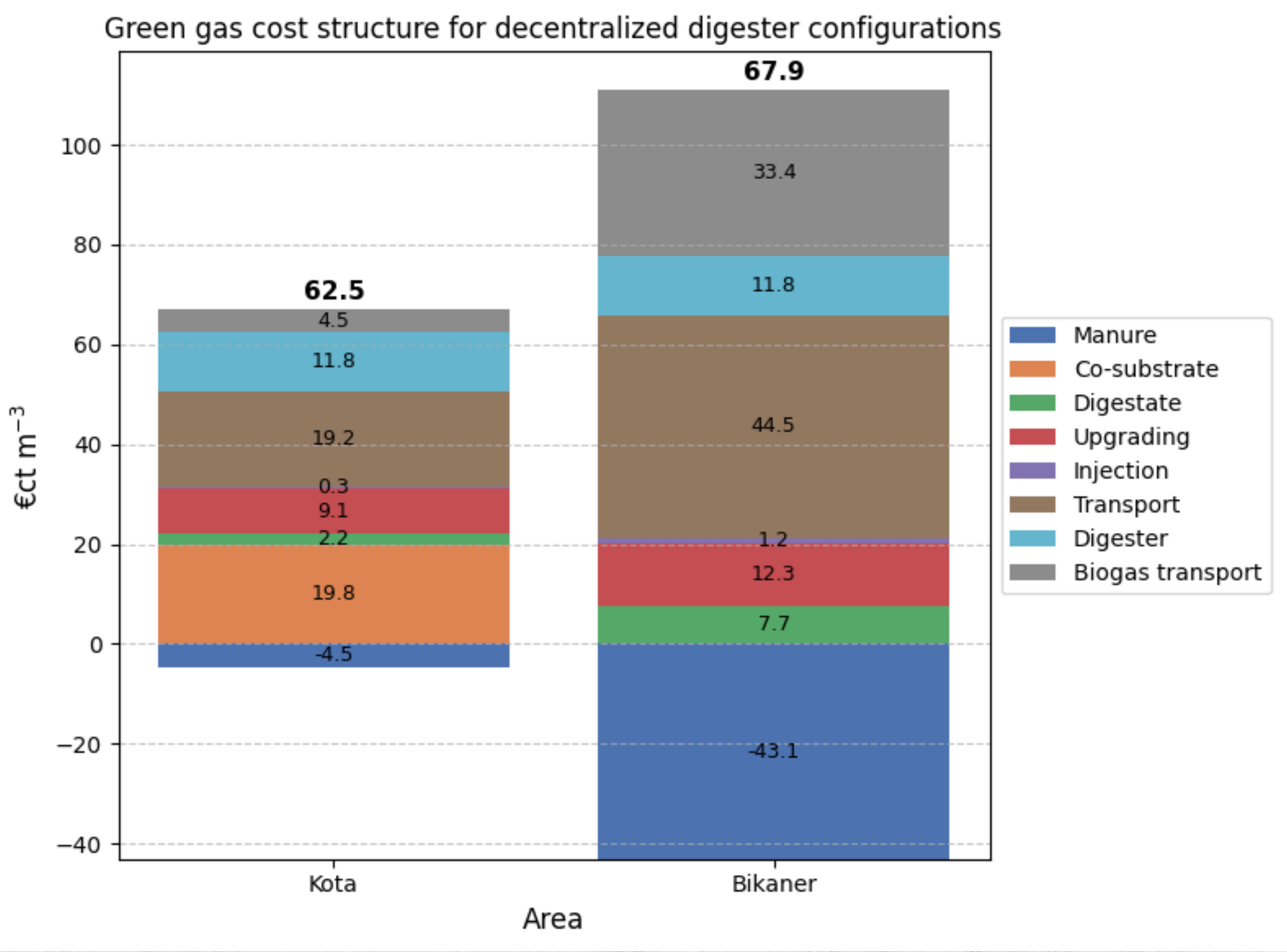
- [1] Number of Cows: https://gopalanapp.rajasthan.gov.in/TotalRegGaushal_Detail_1960.aspx
- [2] Maize Production: <https://www.scribd.com/document/726510619/RACP-ABPF-Tech-DPR-Maize-Flakes>
- [3] Production Area: <https://rajas.rajasthan.gov.in/PDF/11222024122534PMAgriculturalStatistics.pdf>
- [4] Biogas Production: <https://blog.anaerobic-digestion.com/how-much-biogas/>

Results – Cost Calculation

Microsoft Excel computational model was employed to calculate Biogas distribution network parameters and infrastructure investment in Kota and Bikaner.

The total production cost for Kota and Bikaner is **62.5 €ct/m³** and **67.9 €ct/m³**.

The cost analysis reveals that Bikaner's production expenses are **8.6%** higher than Kota.

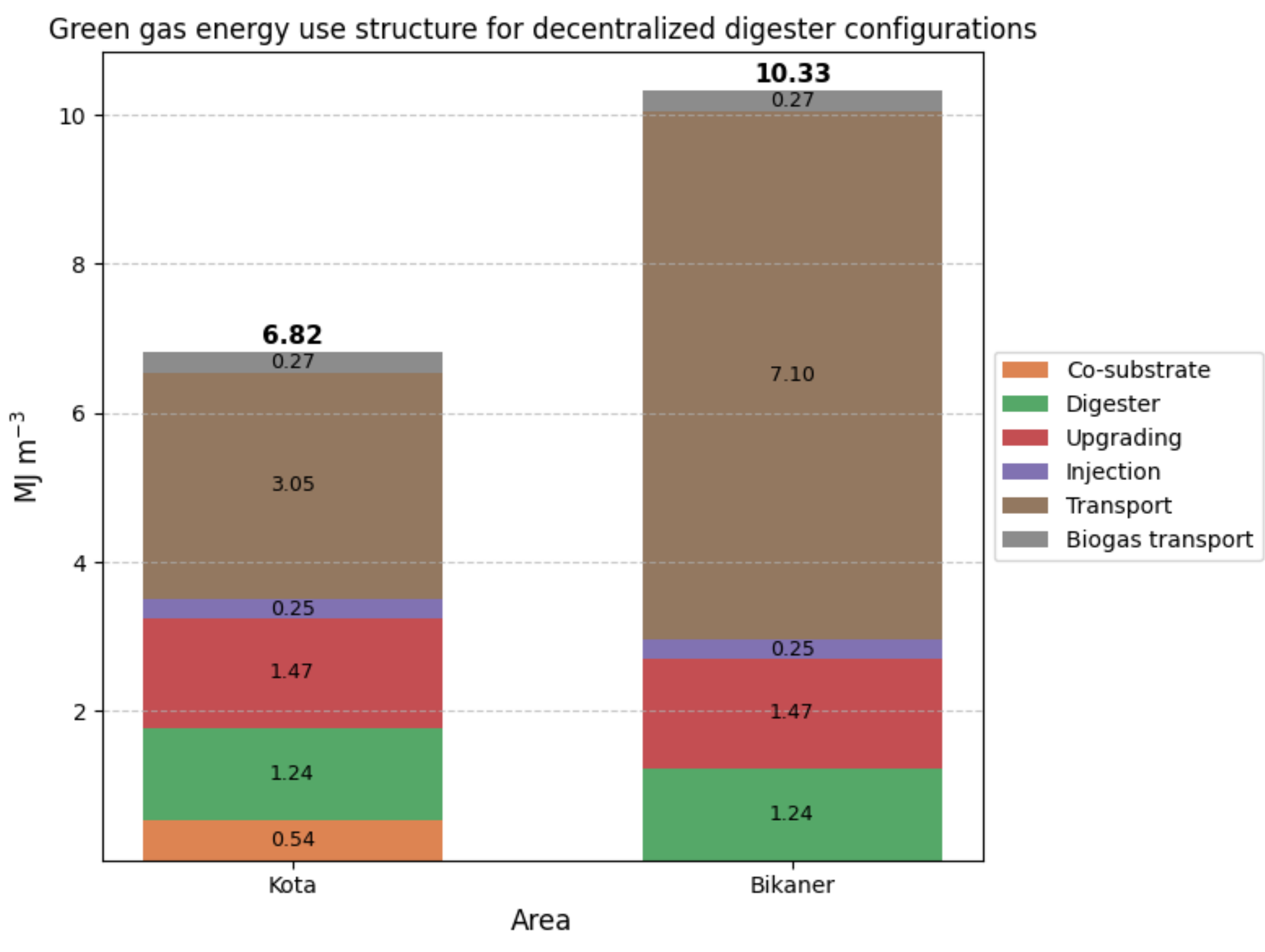


Results – Energy Calculation

The total energy consumption for Kota is **6.82 MJ/m³**, and for Bikaner is **10.33 MJ/m³**.

The primary energy consumers in both locations are transport, upgrading and digester operation processes.

The energy analysis reveals that Bikaner is 51.5% more energy-intensive than Kota for green gas production..



Challenges Faced

- Difficulty in selecting an appropriate and representative location for the analysis.
- Lack of reliable and consistent parametric data from literature.
- Uncertainty in key input parameters such as number of cows, available area, biomass solid content, and volatile solids content.
- Variability in conversion efficiencies from biomass to biogas and from biogas to green gas.
- Challenges in estimating the optimal distance between the biogas plant and the gas grid injection point.

Conclusion

Which location is better for a Biogas Plant?

KOTA

BIKANER

Based on our excel model Kota is more suitable location than Bikaner.

Lower Production Cost

Better Energy Efficiency

Lower Transport Costs

**Reduced Biogas
Transport**

**Flexible Feedstock
Management**

Gantt Chart

